

# **EC442: Lecture Notes**

Chen Gao

Notes on the 1st year Macro class

September 29, 2026

# Contents

|                                                  |   |
|--------------------------------------------------|---|
| 1. Lecture 1 Sep. 29 .....                       | 3 |
| 1.1. Why heterogeneity .....                     | 3 |
| 1.2. How to model incomplete markets? .....      | 3 |
| 1.3. Planning problem .....                      | 3 |
| 1.4. Towards a recursive formulation .....       | 4 |
| 1.5. Dynamic Programming .....                   | 5 |
| 1.6. Contraction mapping theorem .....           | 5 |
| 1.6.1. Application to planner .....              | 5 |
| 1.7. Using the value function analytically ..... | 6 |
| 1.8. Using the value function numerically .....  | 6 |
| Bibliography .....                               | 8 |

# 1. Lecture 1 Sep. 29

## 1.1. Why heterogeneity

Complete markets + standard welfare assumptions:

- Competitive eq. Pareto efficient; planner representation w/ fixed weights  $\alpha_i > 0$ .
- Common discounting; preferences  $u(c) - v(n)$ . Consumption FOCs:

$$\alpha_i u'(c_t^i) = \alpha_j u'(c_t^j) \Rightarrow \frac{u'(c_t^i)}{u'(c_t^j)} = \frac{\alpha_j}{\alpha_i}. \quad (1.1)$$

**Marginal-utility ratio** constant across time/states.

- Common CRRA:  $u'(c) = c^{-\sigma}$ .

$$\frac{c_t^i}{c_t^j} = \left( \frac{\alpha_i}{\alpha_j} \right)^{\frac{1}{\sigma}}. \quad (1.2)$$

Fixed consumption shares  $\rightarrow$  **perfect insurance** against idiosyncratic income shocks.  
Aggregate risk remains; unequal Pareto weights  $\rightarrow$  unequal consumption.

- Strictly convex labor disutility; interior labor choice; aggregate conditions fixed: productivity  $\uparrow \rightarrow$  labor  $\uparrow$ ; consumption share unchanged.

Implications:

- Karl Marx
- Full insurance at odds w/ data

## 1.2. How to model incomplete markets?

1. **Exogenous:** assume some markets absent.
  1. e.g. save/borrow via noncontingent bonds; no state-contingent claims.
2. **Endogenous:** derive market incompleteness from frictions.
  1. e.g. limited commitment (participation constraints), private information.
  2. deeper but harder to do

## 1.3. Planning problem

Set  $n_t = A = 1$ ;  $f(k) = F(k, 1)$ . Given  $k_0 \geq 0$ . Discount factor  $0 < \beta < 1$ ; depreciation  $0 \leq \delta \leq 1$ .

Choose sequences  $c_t, k_{t+1} \geq 0$ :

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \text{ s.t. } c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t. \quad (1.3)$$

- **Lagrangian**; current-value multipliers  $\lambda_t$ :

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (f(k_t) + (1 - \delta)k_t - c_t - k_{t+1})]. \quad (1.4)$$

- **Interior FOCs:**

$$\begin{aligned} c_t : u'(c_t) &= \lambda_t, \\ k_{t+1} : \lambda_t &= \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]. \end{aligned} \quad (1.5)$$

Euler equation:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta]. \quad (1.6)$$

- **Transversality condition:**

$$\lim_{T \rightarrow \infty} \beta^T \lambda_T k_{T+1} = \lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0. \quad (1.7)$$

Discounted shadow value of terminal capital  $\rightarrow 0$ .

- Increasing, concave  $u$ ; concave  $f$ ; finite discounted utility: feasibility + FOCs + TVC  $\rightarrow$  global optimum. At corners, use KKT conditions.

## 1.4. Towards a recursive formulation

- Sequence approach: infinitely many dated choices + optimality conditions.
- Repeated trade-off: consume today vs. save for tomorrow.
- Stationary problem; current capital  $k_t$  summarizes relevant history.
- Recursion: continuation value summarizes future choices; same decision problem at every date.

**Value function:** maximal lifetime utility from initial capital  $k_0$ :

$$v^*(k_0) = \max_{\{c_t, k_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t). \quad (1.8)$$

Feasible sequences:  $c_t, k_{t+1} \geq 0$ ;  $c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$ , all  $t \geq 0$ .

**Split date 0 from continuation:**

$$v^*(k_0) = \max_{c_0, k_1} \left[ u(c_0) + \beta \max_{\{c_t, k_{t+1}\}_{t \geq 1}} \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \right]. \quad (1.9)$$

- Outer max:  $c_0, k_1 \geq 0$ ;  $c_0 + k_1 = f(k_0) + (1 - \delta)k_0$ .
- Inner max: given chosen  $k_1$ , same feasibility constraints for all  $t \geq 1$ . Discounting restarts at date 1  $\rightarrow$  inner value  $v^*(k_1)$ .

Continuation value absorbs future choices. **Bellman equation** (current capital  $k$ , next-period capital  $k'$ ):

$$v^*(k) = \max_{c, k'} [u(c) + \beta v^*(k')] \quad (1.10)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.11)$$

Infinite sequence  $\rightarrow$  one-period choice + discounted continuation value. Stationarity  $\rightarrow$  same problem at every date.

Bellman equation = **functional equation**; unknown = function  $v(k)$ .

Questions:

- **Existence**: conditions for solution  $v$ ?
- **Uniqueness**: conditions for unique  $v$  within chosen function class?
- **Equivalence**: when does  $v = v^*$ , with maximizing policy solving original sequence problem?

Answer:

- *Recursive Macroeconomic Theory* (RMT), “Dynamic Programming” (Ljungqvist & Sargent, 2018, ch. 3).

## 1.5. Dynamic Programming

**Bellman operator**  $T$ : candidate continuation value  $v \rightarrow$  updated value  $Tv$ .

$$(Tv)(k) = \max_{c,k'} [u(c) + \beta v(k')] \quad (1.12)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.13)$$

Bellman solution = **fixed point**:

$$Tv = v \Leftrightarrow (Tv)(k) = v(k) \text{ for all feasible } k. \quad (1.14)$$

Fixed-point theorems give existence/uniqueness conditions within chosen function space; assumptions matter.

## 1.6. Contraction mapping theorem

**Theorem 1.6.1** (Contraction mapping theorem): Let  $(S, d)$  be a nonempty complete metric space and  $T : S \rightarrow S$  a contraction: there exists  $\beta \in (0, 1)$  such that

$$d(Tx, Ty) \leq \beta d(x, y) \text{ for all } x, y \in S. \quad (1.15)$$

Then  $T$  has a unique fixed point  $v \in S$ . For every  $v_0 \in S$  and  $n = 0, 1, 2, \dots$ ,

$$d(T^n v_0, v) \leq \beta^n d(v_0, v). \quad (1.16)$$

Hence  $T^n v_0 \rightarrow v$  as  $n \rightarrow \infty$ , with geometric convergence. Here  $T^n$  denotes  $n$  successive applications of  $T$ , with  $T^0$  the identity.

### 1.6.1. Application to planner

- Assume feasible capital remains in nonempty compact interval  $K$ .
- $S = C(K)$ : continuous real-valued functions on  $K$ ; automatically bounded.
- **Sup-norm metric**:

$$d(v, w) = \|v - w\|_\infty = \sup_{k \in K} |v(k) - w(k)|. \quad (1.17)$$

$(S, d)$  complete.

- Must verify  $T : S \rightarrow S$ : updated values remain bounded and continuous. Then Bellman discounting gives

$$\|Tv - Tw\|_{\infty} \leq \beta \|v - w\|_{\infty}. \quad (1.18)$$

**Theorem 1.6.1**  $\rightarrow$  unique fixed point in  $S$ ; value iteration converges from any  $v_0 \in S$ .

## 1.7. Using the value function analytically

Assume  $u$ ,  $f$ , and  $v$  differentiable; optimal choices interior. Substitute  $c = f(k) + (1 - \delta)k - k'$  into Bellman:

$$v(k) = \max_{k'} [u(f(k) + (1 - \delta)k - k') + \beta v(k')], \quad (1.19)$$

over feasible  $k'$ .

- **FOC for  $k'$ :**

$$-u'(c) + \beta v'(k') = 0 \Rightarrow u'(c) = \beta v'(k'). \quad (1.20)$$

- **Envelope condition:** differentiate w.r.t.  $k$ , holding optimal  $k'$  fixed:

$$v'(k) = u'(c)[f'(k) + 1 - \delta]. \quad (1.21)$$

- **Combine:** apply envelope condition next period  $\rightarrow$  usual Euler equation:

$$u'(c_t) = \beta v'(k_{t+1}) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]. \quad (1.22)$$

Same condition as sequence problem. At corners, use KKT conditions.

## 1.8. Using the value function numerically

Need numerical representation of  $v(k)$ .

- Simplest approach: restrict capital to finite grid

$$K_N = \{k^1, k^2, \dots, k^N\} \subset K. \quad (1.23)$$

- Approximate value function by vector

$$\mathbf{v} = (v_1, v_2, \dots, v_N)^{\top} \in \mathbb{R}^N, v_i \approx v(k^i). \quad (1.24)$$

One entry per capital grid point.

**Algorithm 1.8.1** (Value function iteration (VFI)):

1. **Choose grid**  $K_N = \{k^1, \dots, k^N\}$ . For each pair  $(i, j)$ , compute

$$c_{ij} = f(k^i) + (1 - \delta)k^i - k^j. \quad (1.25)$$

Let  $J_i$  contain choices  $j$  with  $c_{ij} \geq 0$  and finite  $u(c_{ij})$ . Each  $J_i$  must be nonempty.

2. **Initial guess:** any  $\mathbf{v}^0 \in \mathbb{R}^N$ , e.g. all zeros. Set iteration  $m = 0$  and tolerance  $\varepsilon > 0$ .
3. **Bellman update:** for every  $i = 1, \dots, N$ ,

$$v_i^{m+1} = \max_{j \in J_i} [u(c_{ij}) + \beta v_j^m]. \quad (1.26)$$

Use the old vector for all updates:  $\mathbf{v}^{m+1} = T_N \mathbf{v}^m$ .

4. **Convergence check:**

$$\Delta_m = \max_{1 \leq i \leq N} |v_i^{m+1} - v_i^m|. \quad (1.27)$$

If  $\Delta_m \leq \varepsilon$ , stop and return  $\hat{\mathbf{v}} = \mathbf{v}^{m+1}$ . Otherwise set  $m \leftarrow m + 1$  and repeat step 3.

5. **Recover approximate policy:** for each  $i$ , choose

$$j_i \in \arg \max_{j \in J_i} [u(c_{ij}) + \beta \hat{v}_j]. \quad (1.28)$$

Set  $k'(k^i) = k^{j_i}$  and  $c(k^i) = c_{ij_i}$ .

**Why it converges:**  $T_N$  is a  $\beta$ -contraction in the sup norm. **Theorem 1.6.1**  $\rightarrow$  unique grid fixed point  $\mathbf{v}_N^*$ ; **Algorithm 1.8.1** converges from any initial vector. At stopping, value-error bound:

$$\|\hat{\mathbf{v}} - \mathbf{v}_N^*\|_\infty \leq \frac{\beta}{1 - \beta} \Delta_m \leq \frac{\beta}{1 - \beta} \varepsilon. \quad (1.29)$$

This controls iteration error; grid approximation error remains.

## Bibliography

Ljungqvist, L., & Sargent, T. J. (2018). *Recursive Macroeconomic Theory* (4th ed.). MIT Press. <https://mitpress.mit.edu/9780262038669/recursive-macroeconomic-theory/><sup>o</sup>